

Shashank Yadav Kalki

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PROFESSIONAL SUMMARY

Machine Learning postgraduate with strong foundations in AI, statistical modelling, dynamical systems, and Generative AI applications. Experienced in building end-to-end ML pipelines, Retrieval-Augmented Generation (RAG) systems, and analysing model behaviour, reliability, and failure modes. Interested in applying machine learning, LLM-powered systems, and data-driven solutions in real-world enterprise environments.

TECHNICAL SKILLS

Machine Learning: Regression, Classification, Clustering, Dimensionality Reduction, Time Series Analysis
Deep Learning: Neural Networks, CNNs, RNN/LSTM, Model Optimization
Generative AI: Retrieval-Augmented Generation (RAG), Embeddings, Prompt Engineering, Vector Search, LLM Evaluation
LLM Frameworks: LangChain, LangGraph, OpenAI API, ChromaDB, FAISS
AI Systems: Multi-Agent Workflows, Tool Calling, Agent Memory, Agent Orchestration
Programming: Python (NumPy, pandas, scikit-learn), R
Data Engineering: Data Cleaning, Feature Engineering, Validation, Bias & Error Analysis
Tools & Platforms: SQL, Git/GitHub, FastAPI, Streamlit, Docker, AWS
Cloud & Deployment: AWS, Docker, FastAPI, REST APIs

PROJECTS

Confabulation Dynamics in Reservoir Computing – Research Project May 2025 – Sept 2025

University College Cork / Under Dr. Andrew Flynn

- Studied how reservoir computing systems generate “confabulations” (plausible but incorrect outputs).
- Analyzed dynamical systems behavior and emergence of untrained attractors.
- Investigated model reliability and failure modes in recurrent neural networks.
- Connected concepts from nonlinear dynamics with machine learning system behavior.

LLM-Powered Document Intelligence Platform Jan 2025 – Feb 2025

Python / FastAPI / OpenAI API / ChromaDB

- Developed an AI-powered document analysis platform for extracting, summarizing, and querying information from unstructured documents.
- Built document ingestion and embedding pipelines supporting semantic retrieval workflows.
- Exposed AI services through REST APIs enabling integration with external applications.
- Implemented response validation and prompt engineering strategies to improve reliability and consistency.

Multi-Agent Research Assistant Aug 2025 – Sept 2025

Python / LangGraph / LangChain / OpenAI API / ChromaDB

- Developed a multi-agent AI system capable of coordinating specialized agents for retrieval, summarization, and response generation tasks.
- Implemented agent orchestration workflows using LangGraph to enable task decomposition and collaboration between agents.
- Integrated tool-calling capabilities allowing agents to access external knowledge sources and APIs.
- Designed short-term memory mechanisms to maintain context across multi-step interactions.
- Evaluated system performance for response quality, task completion, and reliability across diverse research queries.

Conversational AI Assistant for Knowledge Retrieval Sept 2024 – Nov 2024

Python / LangChain / OpenAI API / ChromaDB / FastAPI

- Built an end-to-end Retrieval-Augmented Generation (RAG) system supporting semantic search across 10,000+ documents.
- Implemented document ingestion and embedding generation pipeline reducing retrieval latency by 30% .
- Integrated OpenAI APIs with FastAPI services to deliver grounded and context-aware responses.
- Applied prompt engineering and response-grounding techniques to improve answer relevance and reduce hallucinations.
- Evaluated retrieval quality, response accuracy, and system reliability across multiple document sources.

EDUCATION

University College Cork Cork, Ireland

M.Sc. Mathematical Modelling & Machine Learning (Second Class Honours Grade I) 2024 Sept – 2026 March

Mahatma Gandhi Institute of Technology Hyderabad, India

B.Tech Computer Science (Data Science) 2020 – 2024

CERTIFICATIONS

- Data Science for Engineers – IIT Madras
- Natural Language Processing – IIT Kharagpur
- Data Analysis with Python – FreeCodeCamp
- Social Network Analysis – IIT Ropar
- Prompt Engineering with OpenAI – Columbia University
- Generative AI Masterclass – Udemy